1. Identity

Document	STAR-PKG-STAR-POLAR-EN · v2.2 · 2026-11-02
Package	star-polar
Version	0.1.0
Capability	star.geodesy.polar-stereographic
Maturity	Released & Verified (RELEASED)
Python	⩾3.9
Source commit	dc09f61
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_polar

2. Licence and obligations

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3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_polar-0.1.0-py3-none-any.whl
```

FILE	SHA-256
star_polar-0.1.0-py3-none-any.whl	1db49e6654cca0541ce0f325165e7b7adcea703694eee6ca1ea62ff630a1dc45
star_polar.py	bb9766d610909103690216cf054a71c10459517903d5bc66a254df2f1347afff
star_polar-0.1.0-source.zip	05869c9363600faeb26f7fd2916a3206a71e83c392ac321ba801e5285e510c90

5. What it does, and its limits

Polar stereographic projection at either pole with any parallel of true scale, forward, inverse and scale factor, and the UPS grid; within 7e-8 m of PROJ and 5e-9 m of PyGeodesy on 600 cases; reproduces Snyder's published example and refuses out-of-range input.

Polar aspect only: no oblique or equatorial stereographic. `ups_forward` does not restrict the latitude to the UPS zones (north of 84 N, south of 80 S): it is the caller's choice of grid. No MGRS letters. PROJ's `+proj=ups` differs from PyGeodesy and from this module by up to 4.4e-5 m while PROJ's `+proj=stere` agrees to 7e-8 m: recorded as a candidate observation about a third-party library, not investigated and not reported.

- `ps_forward`, `ps_inverse`: Snyder's formulas (USGS PP 1395, 21-33 ... 21-41) for either pole, with the parallel of true scale given by the caller (or the pole) and a scale multiplier k0; the isometric latitude is written without cancellation. Reproduce Snyder's numerical example to the printed 0.1 m.
- `ps_scale`: point scale factor, exactly 1 on the parallel of true scale. `ups_forward`, `ups_inverse`: UPS on WGS-84 (k0 = 0.994, 2,000,000 m false easting and northing).
- Measured on 600 cases at both poles (latitudes 20 to 89.999 deg, true scale at 50 to 90 deg): forward within 7e-8 m and inverse within 5e-8 m of PROJ, scale within 6e-11; UPS within 5e-9 m of PyGeodesy (forward and inverse) and 4.4e-5 m of PROJ's `ups`.
- Every function returns finite floats or raises `ValueError`: non-numeric, boolean, NaN or infinite input, a pole other than 'N' or 'S', a latitude more than 60 deg beyond the equator, a parallel of true scale in the other hemisphere or within 1 deg of the equator, an ellipsoid or scale outside the stated ranges.

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_polar_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-polar}: Polar stereographic projection and UPS},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_polar/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit dc09f61}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

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